

RESEARCH ARTICLE

Decision Accountability in LLM-Driven UAVs: A Privacy-Preserving Forensic Logging Framework

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ABSTRACT Large language models (LLMs) are increasingly being integrated into unmanned aerial vehicle (UAV) systems to support mission planning, policy interpretation, safety assessment, and cyber-physical decision making. However, decisions generated by LLM-guided UAVs are often ephemeral, while conventional UAV logs mainly preserve telemetry, sensor measurements, control traces, and command records. These artifacts do not ordinarily preserve a structured and verifiable record linking a discrete decision to its retained context, selected action, temporal position, and integrity evidence. This limits accountability, forensic traceability, and regulatory auditability in privacy-sensitive operational environments. This paper proposes a privacy-preserving forensic logging framework for LLM-driven UAV decision systems. The framework introduces a compact DecisionRecord abstraction that captures decision semantics, temporal ordering, and integrity metadata in a model-interface-agnostic form. It also defines a constrained DecisionRecord' representation under the P2 profile, excluding raw prompts, unrestricted reasoning, telemetry streams, precise locations, and complete candidate-choice lists while retaining selected evidence for post-incident investigation. Privacy preservation is operationalized as bounded data minimization rather than anonymization, PII detection, formal privacy protection, or resistance to inferential disclosure. Evidentiary integrity is protected using deterministic canonical serialization, SHA-256 hash chaining, and trusted checkpoint comparison. Evaluation using 200 deterministically sampled UAVBench instances produced 0/800 whole-field verbatim retention for the evaluated omitted source fields, 1,400/1,400 predefined evidence-group availability, and a 49.224% reduction in average canonical payload size relative to the canonically serialized source MCQs. Local verification detected all 20/20 independently executed record modifications and all 5/5 interior deletions. All five coherently recomputed suffixes passed local verification but were detected through mismatch with an independently retained trusted terminal checkpoint. Across 30 workstation executions, mean end-to-end record-construction latency was 0.563214 ms per record, and the complete 200-record chain occupied 220.795 KiB. The results establish benchmark-grounded feasibility for compact, data-minimized, and tamper-evident preservation of discrete LLM-guided UAV decision events. They do not establish decision accuracy, authenticated model identity, recovery of hidden reasoning, complete mission reconstruction, or onboard UAV performance.

INDEX TERMS Autonomous systems, cyber-physical security, decision accountability, forensic logging, large language models, privacy preservation, unmanned aerial vehicles.

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I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are increasingly incorporating large language models (LLMs) into mission-level autonomy, where they support high-level planning, policy interpretation, safety assessment, operator interaction,

and decision-making under uncertainty. This transition is significant because UAV autonomy is no longer limited to perception, control, and rule-based execution; it increasingly involves semantic reasoning over operational context, constraints, and goals. As a result, LLM outputs may influence actions whose consequences are safety-critical, compliance-sensitive, and operationally consequential.

Despite this growing reliance on LLM-guided autonomy, current UAV systems provide limited support for preserving decision-level forensic evidence. Existing logging mechanisms largely emphasize telemetry, sensor measurements, communication traces, and control commands. Although these artifacts are essential for reconstructing vehicle motion and physical system state, they do not ordinarily preserve a structured record linking a discrete LLM-guided decision to its retained context, selected action, declared model metadata, temporal position, and integrity evidence. Investigators may therefore observe what the UAV executed without having a compact and verifiable artifact of the recorded decision event. This gap complicates forensic investigation, accountability, compliance assessment, and failure analysis.

Benchmark-based accuracy evaluation does not resolve this forensic gap. Recent benchmarking efforts have substantially improved the evaluation of LLM reasoning in UAV-related tasks. Benchmarks such as UAVBench [1] assess decision quality across diverse operational scenarios and reasoning domains, thereby advancing measurement of correctness, robustness, and domain competence. However, such benchmarks are primarily designed for performance evaluation rather than forensic preservation. They assess whether a model selects a correct or plausible answer, but they do not define how the selected output, retained event context, declared model metadata, temporal position, and integrity evidence should be represented and preserved as a forensic artifact. Consequently, both correct and incorrect decisions may remain unavailable for later integrity verification, auditing, incident response, or regulatory review.

Preserving decision semantics, however, introduces a second challenge: the resulting evidence may contain privacy- and security-sensitive operational information. LLM-generated outputs, intermediate rationales, or associated prompts may encode sensitive mission objectives, protected locations, internal policies, operational constraints, or user-specific information. Naively preserving such material may violate confidentiality requirements or expose mission-critical details. Conversely, avoiding decision-level logging altogether limits accountability and later investigation of recorded decision events. The central challenge is therefore not merely to log more data, but to retain bounded decision-level evidence for forensic analysis while minimizing disclosure of operationally sensitive information.

Even a privacy-minimized record cannot serve as reliable forensic evidence unless its integrity and ordering can

be verified. Although tamper-evident logging and audit-chain mechanisms have been studied in other cyber-physical and security settings, they have not been systematically adapted to the semantic decision outputs of LLM-guided UAV systems. This leaves an important gap between the growing use of LLMs in autonomous aerial decision making and the mechanisms needed to support trustworthy forensic accountability.

Accordingly, the problem addressed in this work is how to preserve a verifiable record of an LLM-guided UAV decision while minimizing unnecessary retention of sensitive operational information.

The aim of this study is to determine whether a discrete LLM-guided UAV decision can be transformed into a compact, data-minimized, and tamper-evident forensic artifact. The artifact is intended to support recovery of the recorded event sequence, interpretation of retained decision context, documentation of declared model metadata, and integrity verification without storing raw prompts or unrestricted model reasoning.

To address this aim, the framework defines a conceptual DecisionRecord and its constrained P2 projection, DecisionRecord'. Only DecisionRecord' is stored and evaluated. For each decision event, the evaluated UAVBench adapter deterministically constructs one DecisionRecord', serializes it canonically, computes its SHA-256 digest, and appends it to a tamper-evident hash chain.

The main contributions of this work are as follows:

- A model-interface-agnostic DecisionRecord schema together with a UAVBench-specific deterministic adapter for constructing its stored P2 projection from multiple-choice decision outputs and benchmark metadata.
- A P2 data-minimization profile that omits specified complete source fields, including raw prompts, unrestricted reasoning, telemetry streams, precise locations, and the complete candidate-choice list, while retaining selected evidence needed for forensic interpretation. P2 is explicitly bounded as a data-minimization profile rather than a formal anonymization, PII-removal, or inferential-privacy guarantee.
- A precisely specified tamper-evident mechanism based on canonical UTF-8 JSON serialization and SHA-256 hash chaining, supplemented by trusted checkpoint comparison for detecting coherently recomputed suffixes.
- A reproducible benchmark-grounded evaluation using 200 UAVBench decision outputs, including 30 repeated runtime executions, independent modification and deletion trials, suffix-recomputation trials, storage and residual-information audits, ablation analysis, and scalability projections.

UAVBench is used as a reproducible source of independent discrete decision events. The benchmark is not modified, and no model training or fine-tuning is performed. The evaluation examines deterministic forensic record construction, data

minimization, ordering, integrity verification, storage, and workstation processing cost. It does not evaluate LLM reasoning accuracy, authenticate the declared model identity, reconstruct a coherent operational mission timeline, or establish software-in-the-loop, hardware-in-the-loop, onboard, or physical-UAV performance.

The remainder of this paper is organized as follows. Section II reviews the related literature. Section III defines the conceptual DecisionRecord. Section IV presents the privacy-preserving DecisionRecord' design. Section V specifies the integrity mechanism and threat model. Section VI describes the experimental setup, and Section VII reports the results. Section VIII discusses the findings, limitations, and future work, while Section IX concludes the paper.

II. RELATED WORK

The literature relevant to this study is organized into three categories: LLM-driven UAV autonomy, UAV forensics and incident analysis, and IoD security, privacy, and integrity. Within each category, the following discussion examines the principal method and application context, the capability provided, the limitation relevant to decision-level forensic accountability, and the evaluation or data-utility measure used by the cited work. This structure clarifies how existing approaches address individual parts of the problem but do not jointly preserve privacy-minimized decision semantics and tamper-evident integrity.

A. LLM-DRIVEN AUTONOMY IN UAV SYSTEMS

The integration of LLMs into robotic systems has shifted the focus from traditional rule-based control to data-driven semantic reasoning [2], [3]. In the context of UAVs, LLMs are increasingly employed to handle high-level mission planning, interpret natural language commands, and manage complex environmental constraints [4]. For instance, the NeLV system demonstrates a five-level automation taxonomy for LLM-as-Pilot architectures [4], while recent works have integrated Retrieval-Augmented Generation (RAG) to enhance decision reliability in safety-critical IoD operations [5].

Despite these functional advancements, evaluating the *reasoning* integrity of these models remains a challenge. Benchmarking efforts such as *UAVBench* [1] and *UAVThreat-Bench* [6] have been introduced to assess decision accuracy and cybersecurity awareness in LLM-driven UAV systems. However, these benchmarks primarily focus on performance metrics—such as success rates in obstacle avoidance or threat identification—rather than the forensic preservation of the decision process itself [7]. As noted by Hu et al., ensuring robustness against hallucinations and adversarial inputs in multi-agent systems requires rigorous certification mechanisms that are currently absent in standard logging pipelines [8].

B. UAV FORENSICS AND INCIDENT ANALYSIS

Digital forensics for UAVs has traditionally focused on the extraction and analysis of low-level telemetry and physical

storage artifacts [9]. Foundational forensic protocols require the recovery of artifacts from both on-board flash memory and external ground control stations [10]. Once acquired, standard procedures involve parsing proprietary logs, such as .DAT and .TXT files, to reconstruct motor states and sensor readings [11]. The complexity of UAV incident response is increased by the high volume of data exchanged between sensors and actuators, necessitating a 'purple-teaming' approach to evaluate the integrity of both static and live digital evidence [12].

Recent literature has expanded towards "forensic-by-design" frameworks; specifically, the Enhanced UAV Forensic Framework (EUAVFF) [13] and the Conceptual Drone Forensic Framework (CDFF) [14] propose modular approaches to handle the heterogeneity of drone data. Further innovations include the use of Digital Twins to simulate and validate accident scenarios based on recovered logs [15]. As UAV systems increasingly rely on LLM-guided decision components, traditional telemetry logs do not ordinarily preserve a dedicated record linking the selected action to its retained decision context; this mirrors the broader crisis in AI where post-hoc explanations of black-box models are increasingly viewed as insufficient for high-stakes decision-making [16].

C. SECURITY, PRIVACY, AND INTEGRITY IN IoD

The deployment of UAVs in sensitive environments necessitates stringent security and privacy controls [17]. The "Internet of Drones" (IoD) paradigm faces unique threats, including GNSS spoofing, man-in-the-middle attacks, and unauthorized data interception [18], [19]. To mitigate these, Zero Trust Architectures (ZTA) have been proposed to enforce continuous authentication and least-privilege access across drone swarms [20], [21].

Privacy preservation is particularly critical when logging operational data. Frameworks such as SP-IoUAV employ federated learning and differential privacy to detect intrusions without exposing raw sensor streams [22]. Similarly, the SHE framework leverages homomorphic encryption to secure data collection in third-party UAV services [23].

Integrity of the logs themselves is a parallel requirement. Tamper-evident logging, often implemented through cryptographic hash chains or external anchors, supports detection of unauthorized post-commitment modification when an uncompromised integrity reference remains available [24]. While lightweight cryptographic mechanisms have been optimized for resource-constrained drones [25], they have not been systematically applied to the semantic decision records of LLM-guided UAV systems. The proposed framework addresses this gap by applying tamper-evident integrity protection to bounded, data-minimized decision artifacts.

Table 1 synthesizes the reviewed literature using a common set of criteria. Because the studies address different problems, their evaluation measures are heterogeneous. The final column therefore reports the principal utility or evaluation

focus used in each line of work rather than implying that all studies share a directly comparable numerical metric.

The comparison reveals three complementary gaps. LLM and benchmark studies evaluate generated decisions but generally do not preserve decision-level forensic artifacts. UAV-forensic approaches reconstruct physical and telemetry events but do not ordinarily represent a discrete LLM-guided decision through a dedicated structured record. Privacy and cryptographic approaches protect data disclosure or integrity but do not determine which decision-event evidence should be retained for later investigation. The proposed framework addresses the intersection of these gaps through deterministic record construction, bounded data minimization, and tamper-evident ordered logging. Its evaluation therefore measures retained evidence availability, residual information, integrity-attack detection, storage, and workstation processing cost.

III. FORENSIC DECISION RECORD DEFINITION

This section formalizes the notion of a *forensic decision record* for large language model (LLM)-driven unmanned aerial vehicle (UAV) systems. We first define decision events in the context of LLM-guided autonomy, then introduce the *DecisionRecord* as a minimal yet sufficient forensic artifact, and finally explain how this representation maps naturally to benchmark-based UAV decision instances.

A. DECISION EVENTS IN LLM-GUIDED UAV SYSTEMS

Recent advances in LLMs have enabled UAV systems to perform high-level reasoning tasks such as mission planning, policy compliance assessment, safety-critical decision making, and cyber-physical threat evaluation. In these settings, the LLM is typically invoked at discrete decision points to evaluate an abstracted operational context and recommend or select an action from a bounded set of alternatives. We refer to each such invocation as a *decision event*.

Although UAV operation is fundamentally continuous, decision-making in practical autonomous architectures is often organized around discrete triggers generated by the broader control stack. Such triggers may arise from time-based decision intervals, mission-state transitions, spatial threshold crossings, anomaly detection events, operator queries, or policy-relevant environmental changes. The continuous lower-level flight dynamics therefore remain outside the *DecisionRecord* itself; instead, the record captures the semantically meaningful decision point at which the system evaluates context and commits to a higher-level choice. This abstraction is consistent with how LLM-mediated autonomy is commonly integrated into hierarchical UAV systems, where continuous control and discrete reasoning operate at different layers.

A decision event comprises three essential components:

- (i) a decision context describing the relevant operational situation and constraints,

- (ii) a decision output corresponding to the selected action or option, and
- (iii) a rationale, either explicit or implicit, that explains why the selected action was preferred over available alternatives.

Conventional UAV logging mechanisms capture low-level telemetry, sensor measurements, and control commands, but they do not explicitly preserve semantic decision-level information such as abstract context, decision influences, or structured justification. As a result, post-incident analysis may reveal what action occurred without adequately explaining why that action was produced. For systems in which LLM outputs influence safety-critical behavior, this omission creates a significant gap in accountability, attribution, and forensic reconstruction.

From a forensic perspective, decision events therefore constitute critical points of evidence. When an incident occurs, investigators must be able to reconstruct not only the action taken, but also the abstract context, constraints, and reasoning signals under which the action was produced. This requirement motivates a dedicated forensic artifact that treats LLM-guided decisions as first-class evidence rather than as transient intermediate outputs.

B. *DecisionRecord*: STRUCTURE AND SEMANTICS

To address this gap, we introduce the *DecisionRecord*, a structured and persistent representation of a single LLM-guided UAV decision event. The *DecisionRecord* is designed to be model-agnostic, scenario-agnostic, and independent of any particular UAV platform, simulator, or deployment environment. Its purpose is to support post-incident reconstruction, attribution, and integrity verification of decision semantics while remaining compatible with privacy-preserving logging deployments.

Formally, let a *DecisionRecord* be defined as

$$DR = \langle v, d, C, M, O, T, I \rangle,$$

where v denotes the record schema version, d the unique decision identifier, C the abstract context reference, M the LLM metadata, O the decision output, T the temporal metadata, and I the integrity metadata. This representation is intentionally compact: it preserves the minimum information required for forensic interpretation without reproducing full prompts, raw contextual inputs, or verbose free-text reasoning.

The context component is defined as

$$C = \langle s, b, r \rangle,$$

where s denotes the scenario class, b the benchmark or instance identifier, and r the dominant reasoning domain associated with the decision. This design allows the decision event to be linked to its originating operational setting while avoiding direct duplication of sensitive contextual content.

The model metadata component is defined as

$$M = \langle m_n, m_v \rangle,$$

TABLE 1. Comparative synthesis of the principal related-work categories.

Category and representative works	Method and application	Principal strength	Limitation relative to this study	Utility or evaluation focus
LLM-enabled robotics [2], [3]	Prompt-based reasoning, planning, and tool use in robotic or autonomous systems.	Demonstrates that LLMs can support semantic planning and interaction.	Does not define privacy-aware, persistent, or tamper-evident forensic records for UAV decisions.	Task completion, planning success, or qualitative autonomous behavior.
LLM-guided UAV autonomy [4], [5]	LLM-as-pilot automation levels and retrieval-augmented reasoning for UAV or IoD missions.	Improves high-level mission interpretation and access to relevant knowledge.	Focuses on decision generation and reliability rather than post-incident preservation and verification.	Reasoning quality, mission-task performance, or decision reliability.
UAV reasoning and threat benchmarks [1], [6]	Benchmark questions and threat scenarios for evaluating UAV-oriented LLM reasoning.	Provides reproducible and diverse decision and cyber-security scenarios.	Evaluates answer performance but does not preserve each decision as a privacy-constrained forensic artifact.	Decision accuracy, task success, or threat-identification performance.
Traditional UAV forensics [9]–[12]	Acquisition and parsing of on-board storage, ground-station artifacts, telemetry, and proprietary flight logs.	Supports reconstruction of physical state, flight behavior, and system events.	Does not normally preserve a dedicated forensic artifact linking an LLM-guided decision to its selected context, declared model metadata, selected action, temporal position, and integrity evidence.	Artifact recovery, log extraction, event timelines, and telemetry reconstruction.
Forensic-by-design and digital-twin approaches [13]–[15]	Conceptual forensic frameworks and simulation-based reconstruction of UAV incidents.	Provides systematic evidence-handling processes and scenario reconstruction.	Does not jointly specify deterministic LLM decision records, P2 minimization, and record-level cryptographic chaining.	Evidence-source coverage, process completeness, or scenario-reconstruction fidelity.
IoD security and Zero Trust [17]–[21]	Authentication, least-privilege access, intrusion detection, and protection against communication and navigation attacks.	Strengthens preventive and runtime security controls for UAV networks.	Primarily prevents or detects attacks rather than preserving structured decision-level evidence for post-incident investigation.	Attack-detection performance, authentication effectiveness, or network security measures.
Privacy-preserving UAV analytics [22], [23]	Federated learning, differential privacy, and homomorphic encryption for protected UAV data processing.	Reduces exposure of raw operational or sensor data.	Targets privacy during analytics or collection rather than semantic forensic utility and tamper-evident decision preservation.	Detection utility under privacy protection, privacy parameters, or cryptographic processing cost.
Tamper-evident and lightweight cryptography [24], [25]	Hash chaining, external anchoring, and lightweight cryptographic protection for resource-constrained logs or communications.	Provides efficient integrity verification and modification evidence.	Protects generic records or data streams without defining which LLM decision semantics should be retained under privacy constraints.	Verification correctness, digest or communication overhead, and cryptographic cost.
Proposed framework	Deterministic P2 DecisionRecord' construction with canonical JSON and SHA-256 hash chaining.	Jointly preserves selected decision-event evidence, declared model metadata, temporal ordering, bounded data minimization, and tamper-evident integrity.	The current validation is benchmark-grounded and does not establish on-board or closed-loop UAV performance.	Whole-field verbatim retention, predefined evidence-group availability, residual operational information, payload reduction, attack-trial detection, storage size, and per-record latency.

where m_n is the model name and m_v is the model version. These attributes support attribution, reproducibility, and comparison across model variants without exposing internal states or proprietary implementation details.

The decision output component is defined as

$$O = \langle a, \rho \rangle,$$

where a denotes the selected action or option and ρ denotes a structured representation of the decision rationale. The rationale is preserved in abstract form rather than as unrestricted natural language, thereby supporting forensic interpretation while reducing unnecessary privacy exposure.

The temporal component is defined as

$$T = \langle t, q \rangle,$$

where t is the decision timestamp and q is the sequence number assigned within the log. Together, these fields support chronological reconstruction and ordered replay of decision events during forensic analysis.

Finally, the integrity component is defined as

$$I = \langle h_{i-1}, h_i \rangle,$$

where h_{i-1} is the hash of the preceding record and h_i is the hash of the current record. This chaining mechanism binds

each record to the evolving state of the log and supports the detection of modification, deletion, or reordering. For the first record in the sequence, h_{i-1} corresponds to a securely anchored genesis value, whose synchronization and trust assumptions are described later in Section V-C.

Under this formulation, the DecisionRecord preserves five classes of forensic information: decision identity, abstract contextual linkage, model attribution, semantic decision outcome, and evidentiary integrity. These elements are sufficient to reconstruct what decision was made, under which abstract operational framing, by which model version, at what point in the decision sequence, and whether the resulting record chain remains intact.

A concrete instantiation of this abstraction may be implemented in a structured serialization format such as JSON for interoperability with logging pipelines, verification tools, and benchmark-driven experiments. However, the forensic significance of the DecisionRecord lies in its semantic structure rather than in any particular encoding format.

Taken together, the DecisionRecord defines a minimal forensic unit for preserving semantically meaningful evidence about LLM-guided UAV decisions. Rather than storing full prompts, verbose explanations, or raw contextual inputs,

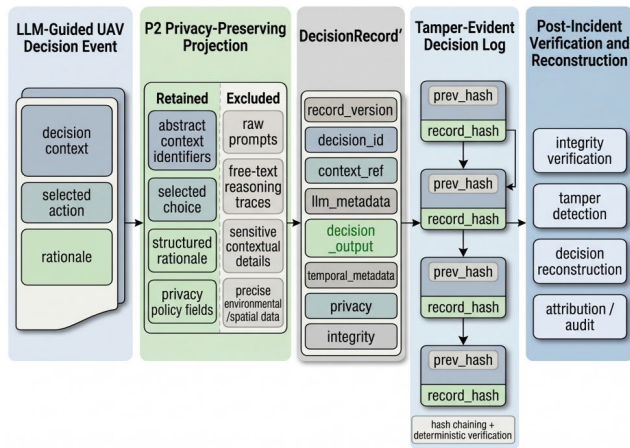


FIGURE 1. Overview of the proposed privacy-preserving forensic logging framework for LLM-driven UAV decisions. LLM-guided decision events are transformed into P2-compliant *DecisionRecord'* entries, linked through tamper-evident hash chaining, and subsequently used for post-incident verification, reconstruction, and attribution.

it records the subset of information necessary for reconstruction, attribution, temporal sequencing, and integrity verification. This balance is central to the framework: the record must remain sufficiently expressive for forensic use while also remaining compact, model-agnostic, and suitable for privacy-sensitive UAV deployments.

Figure 1 summarizes the end-to-end workflow of the proposed framework, highlighting how decision events are abstracted into privacy-preserving forensic records and protected through tamper-evident logging for later post-incident analysis.

IV. PRIVACY-AWARE FORENSIC DESIGN

Forensic accountability requires preserving enough decision semantics to support post-incident reconstruction, attribution, and verification. In LLM-guided UAV systems, however, decision-level artifacts may also contain sensitive operational, spatial, procedural, or policy-related information. A forensic logging mechanism must therefore satisfy two competing requirements: it must retain sufficient semantic structure for investigation, while also limiting disclosure of information that should not be persistently stored or broadly shared. This section analyzes the privacy risks inherent in decision-level logging and introduces a privacy-preserving projection of the *DecisionRecord* that retains forensic utility while minimizing information exposure.

A. PRIVACY RISKS IN LLM-DRIVEN UAV DECISION LOGGING

Unlike traditional UAV telemetry, which is dominated by numeric sensor streams and control signals, LLM-guided decision logs may encode high-level semantic content. Depending on the application, such content may implicitly reveal sensitive mission objectives, protected locations,

operator intentions, internal safety policies, or compliance-related interpretations. Free-text rationales are particularly sensitive because they can expose contextual detail far beyond the minimum evidence needed for post-incident analysis, even when raw sensor data are not retained.

The privacy risk is amplified by the lifecycle of forensic logs. Decision records may be transmitted to ground stations, exported to external investigators, retained for long periods, or shared across organizational and regulatory boundaries. In such settings, storing raw prompts, detailed environmental descriptions, precise spatial references, or unrestricted reasoning traces creates not only confidentiality risk but also downstream compliance and governance concerns. The problem is therefore not simply one of secure storage; it is a representational problem concerning what should be logged in the first place.

These observations motivate a design in which privacy preservation is embedded into the structure of the forensic artifact itself rather than deferred to ad hoc redaction, post-processing, or access-control policies applied after collection.

B. PRIVACY-PRESERVING *DecisionRecord'* (P2 PROFILE)

To address this challenge, the framework defines a privacy-preserving form of the *DecisionRecord*, denoted *DecisionRecord'* under the P2 profile. The purpose of P2 is to preserve the semantically relevant evidence required for forensic accountability while excluding data elements whose retention would create unnecessary privacy or operational risk. In this sense, P2 is not merely a redacted logging format; it is a principled abstraction layer that transforms a richer decision event into a privacy-aware forensic representation.

Formally, let the original *DecisionRecord* be

$$DR = \langle v, d, C, M, O, T, I \rangle,$$

as defined in Section III. Under the P2 profile, this record is projected into a privacy-constrained representation

$$DR' = \langle v, d, C', M, O', T, P, I \rangle,$$

where C' denotes the abstracted context reference, O' the privacy-constrained decision output, and P a declarative privacy-policy component describing the logging configuration under which the record was generated.

The abstracted context component is defined as

$$C' = \langle s, b, r \rangle,$$

where s denotes the scenario class, b the benchmark or instance identifier, and r the dominant reasoning domain associated with the decision. Unlike richer contextual representations, C' excludes raw scenario descriptions, prompt text, sensor observations, and precise environmental references.

The projected decision-output component is defined as

$$O' = \langle a, \rho_{\text{struct}} \rangle,$$

Algorithm 1 Privacy-Preserving Projection to P2 Profile**Require:** Full decision event (context, decision outcome, rationale)**Ensure:** P2-compliant DecisionRecord'

- 1: Initialize empty DecisionRecord'
- 2: Extract and abstract context identifiers (scenario class, benchmark instance, reasoning domain)
- 3: Discard raw prompts and sensitive contextual information
- 4: Map decision outcome to `selected_choice`
- 5: Transform free-text rationale into structured indicators
- 6: Populate P2 privacy policy fields (e.g., `redaction_profile`, `pii_policy`, `prompt_policy`)
- 7: Output populated DecisionRecord'

where a denotes the selected action or option and ρ_{struct} is a structured rationale representation restricted to predefined indicators such as referenced constraints, dominant risk categories, violation classes, and bounded confidence levels. By design, O' excludes unrestricted natural-language explanations and detailed reasoning traces.

The privacy-policy component is defined as

$$P = \langle p_{\text{redact}}, p_{\text{pii}}, p_{\text{prompt}} \rangle,$$

where p_{redact} identifies the redaction profile applied to the record, p_{pii} specifies the treatment of personally identifiable information, and p_{prompt} specifies whether raw prompt content is stored or omitted. Unlike the contextual and decision-output fields, these values do not describe the decision event itself; instead, they document the privacy configuration under which the record was created.

Under this formulation, the P2 profile deliberately excludes raw prompts, unrestricted reasoning traces, precise environmental data, personally identifiable information, and fine-grained spatial references. At the same time, it preserves enough structure to distinguish between materially different forms of decision behavior, such as safety-driven responses, policy-constrained actions, and cyber-physical interference handling. The design objective is therefore not maximal compression, but controlled semantic retention: enough information for forensic sufficiency, but not enough to recreate the full sensitive context of the original decision process.

Algorithm 1 summarizes the high-level procedure by which a full decision event is projected into a P2-compliant DecisionRecord'. The algorithm is intentionally abstract and captures the logical transformation rather than implementation-specific details.

For the evaluated UAVBench adapter, the projection in Algorithm 1 is instantiated deterministically as follows. Let x denote the parsed benchmark instance, f its relative filename, a the archived selected-option label, m the configured model identifier, u the execution identifier, t the assigned UTC timestamp, and q the one-based sequence number. Given the same (x, f, a, m, u, t, q) , the procedure produces the same DecisionRecord'.

The field `record_version` is assigned the constant value 1.0p, and `decision_id` is constructed as DR- u - q ,

with q represented using four digits. Within C' , the scenario field is copied from `x.scenario_name`, the benchmark identifier is derived from the basename of f after removing the literal suffix `_mcq.json`, and the reasoning domain is copied from `x.style`. The configured model identifier m is split at its first slash to obtain `model_name` and `model_version`. If no slash is present, the complete identifier is stored as `model_name`, and unspecified is stored as `model_version` because no separate version component is available.

The archived label a indexes `x.choices` to obtain the selected choice. The selected text is Unicode-NFKC normalized, stripped of any leading option label, converted to uppercase, and transformed into an underscore-separated alphanumeric representation. If no alphanumeric content remains, UNPARSED is stored. To generate ρ_{struct} , the implementation forms a temporary string from the benchmark description, question, and selected-choice text. This string is Unicode-NFKC normalized, converted to lowercase, stripped, and reduced to single spaces. It is used only during indicator generation and is not stored in DecisionRecord'.

The following keyword, style, and character-proximity mappings are implementation rules of the evaluated UAVBench adapter and are not intended as universal forensic semantics. Referenced constraints are generated through fixed keyword rules: ALTITUDE_LIMIT for *altitude* or *agl*; ENERGY_RESERVE for *battery*, *reserve*, *energy*, or *power*; GEOFENCE_COMPLIANCE for *no-fly*, *no fly*, or *geofence*; SEPARATION_REQUIREMENT for *separation*, *collision*, *obstacle*, or *traffic*; NAV_INTEGRITY for *gnss*, *gps*, *navigation*, *jamming*, or *spoofing*; COMMUNICATION_AVAILABILITY for *communication*, *uplink*, *downlink*, *lost-link*, or *lost link*; MISSION_TIME for *deadline*, *mission time*, or *time budget*; and ENVIRONMENTAL_SAFETY for *wind*, *icing*, *visibility*, or *weather*. Duplicate indicators are removed and the retained values are sorted lexicographically.

The dominant risk category is assigned from `x.style_id` according to the following fixed mapping:

- 1: FLIGHT_DYNAMICS
- 2: NAVIGATION_PATH
- 3: POLICY_COMPLIANCE
- 4: ENVIRONMENT_SENSOR
- 5: MULTI_UAV_COORDINATION
- 6: CYBER_PHYSICAL_SECURITY
- 7: ENERGY_RESOURCE
- 8: SAFETY_CRITICAL
- 9: SYSTEM_COMPARISON
- 10: INTEGRATED_MISSION

Violation codes are generated using fixed lexical and character-proximity rules applied to the normalized event text. A GNSS-spoofing indicator is generated when *gps* or *gnss* occurs within a 40-character window of a term beginning with *spoof*, in either order. A GNSS-jamming indicator is generated when *gps* or *gnss* occurs within a 40-character window of a term beginning with *jam*, also in either order.

A communication-loss indicator is generated when *communication*, *uplink*, or *downlink* occurs within a 30-character window of *loss*, *lost*, *outage*, or *failure*, or when the term *lost-link* is present. An icing-condition indicator is generated when *icing* is present, while a wind-shear indicator is generated when the text contains *wind shear*, *wind-shear*, or *windshear*.

A geofence-violation indicator is generated when *geofence*, *no-fly*, or *no fly*, optionally followed by *zone*, occurs within a 35-character window of *violation*, *breach*, *entered*, or *crossed*. A separation-violation indicator is generated when *collision* or *separation* occurs within a 35-character window of *violation*, *breach*, *loss*, or *conflict*.

An energy-reserve-violation indicator is generated when *battery* or *energy* occurs within 35 characters of *reserve*, and *reserve* is followed within 25 characters by *violation*, *breach*, *below*, or *insufficient*. If no rule matches, `NONE_IDENTIFIED` is stored. Duplicate violation codes are removed, and the remaining codes are sorted lexicographically.

Because the archived outputs contain no explicit model-confidence value, `confidence_bucket` is assigned the constant `NOT_REPORTED`. Thus, `$\rho_{\text{struct}}$` contains scenario- and selected-output-derived forensic indicators rather than the model's unrestricted reasoning or an authenticated explanation of its decision. The benchmark `correct_choice` and reference explanation are not used to generate these fields.

The temporal fields are assigned from t and q . The privacy-policy fields are assigned the constants `P2`, `no_raw_pii`, and `no_raw_prompt`. These values document the configured policy and do not independently verify that every retained field is free of sensitive information. Finally, `prev_hash` and `record_hash` are populated by the hash-chain construction defined in Section V-A.

Thus, the P2 profile separates retained forensic semantics, declarative privacy-policy information, and integrity metadata in a mathematically explicit manner. The retained fields are selected because they serve concrete accountability functions: decision identity and temporal metadata support ordering and timeline reconstruction; abstract context references preserve situational linkage without embedding raw operational content; model metadata supports attribution and reproducibility; the structured decision output records both the selected action and an abstracted rationale; and the integrity fields preserve the hash-chain linkage required for later tamper detection.

Equally important are the elements that are intentionally excluded. Raw prompts, unrestricted natural-language rationales, sensor streams, precise locations, and detailed environmental descriptions are omitted because they are not strictly necessary for determining what decision was made, when it occurred, and whether the decision log remains intact. Their exclusion substantially reduces privacy exposure while preserving the core evidentiary value needed for post-incident analysis.

C. EXAMPLE: PRIVACY-PRESERVING *DecisionRecord'* INSTANTIATION

The following synthetic GNSS-spoofing event illustrates the structure of *DecisionRecord'* under the P2 profile.

To illustrate the proposed abstraction concretely, consider a benchmark-driven UAV decision event involving a cyber-physical security scenario in which the UAV must respond to anomalous navigation behavior during mission execution. In the original benchmark instance, the decision context may include rich scenario text, multiple candidate actions, and reasoning cues. Directly logging this material would preserve more context than is necessary for forensic accountability and could expose sensitive operational details.

Under the P2 profile, the logging system records only the abstracted decision semantics required for attribution, reconstruction, and verification. A representative serialized instance of *DecisionRecord'* for such a benchmark-linked event is shown below:

```
DecisionRecord' := {
  record_version: "1.0p",
  decision_id: "DR-2025-0142",
  context_ref {
    scenario_class: "GNSS_SPOOFING_RESPONSE",
    benchmark_instance_id: "UAVBench_MCQ_1023",
    reasoning_style: "cyber_physical_security"
  },
  llm_metadata {
    model_name: "LLaMA-3.2",
    model_version: "vision-instruct"
  },
  decision_output {
    selected_choice: "RETURN_TO_BASE",
    rationale_structured {
      referenced_constraints: ["NAV_INTEGRITY"],
      dominant_risk_category:
        "SIGNAL_MANIPULATION",
      violation_codes: ["GPS_SPOOFING_DETECTED"],
      confidence_bucket: "HIGH"
    }
  },
  temporal_metadata {
    timestamp: "2025-01-14T10:32:45Z",
    sequence_number: 142
  },
  privacy {
    redaction_profile: "P2",
    pii_policy: "no_raw_pii",
    prompt_policy: "no_raw_prompt"
  },
  integrity {
    prev_hash: "a7f9c2e4...",
    record_hash: "e13b8d91..."
  }
}
```

In this example, raw scenario text, unrestricted reasoning traces, sensor measurements, and precise spatial information are intentionally absent. The original context is reduced to an abstract scenario class and a benchmark-linked identifier, allowing investigators to associate the decision with a known class of operational situations without storing the sensitive contents of that situation directly. Similarly, the rationale is represented through structured indicators of risk, constraints, and violations rather than free-text explanation.

This example illustrates the central design principle of the P2 profile: privacy is enforced by construction rather than by later sanitization. Sensitive decision inputs are never written to the forensic log, yet the resulting *DecisionRecord'* remains attributable, temporally ordered, and integrity-verifiable. In this work, all logged and evaluated decision records follow this abstraction model.

D. UTILITY-PRIVACY TRADE-OFFS

The P2 profile represents a deliberately conservative privacy configuration intended for environments with strict confidentiality, governance, or regulatory constraints. This abstraction inevitably reduces the granularity of explanation available to an investigator. However, it does not eliminate forensic utility. Under P2, an analyst can still reconstruct the decision sequence, identify the responsible model version, recover the selected action, and determine which classes of risks, constraints, and violations influenced the outcome.

More permissive profiles could preserve partially redacted free-text rationales or richer contextual references, thereby increasing explanatory depth at the cost of greater disclosure risk. Conversely, more aggressive abstraction would further reduce privacy exposure but may also weaken attribution and interpretability. The key design insight is therefore not that privacy and forensic utility are perfectly aligned, but that they can be balanced through structured representation of decision semantics.

In this work, all experiments adopt the P2 profile to test the framework under a deliberately strict privacy assumption. This choice establishes a lower bound on retained forensic utility: if post-incident attribution, reconstruction, and integrity verification remain feasible under P2, then the framework demonstrates forensic sufficiency even in a conservative privacy setting. Accordingly, the empirical results reported later in the paper should be interpreted as evidence of forensic sufficiency under a privacy-first configuration rather than under a maximally informative logging regime.

V. INTEGRITY PROTECTION AND THREAT MODEL

This section describes the integrity mechanism used to protect forensic decision records from post-incident manipulation and formalizes the threat model under which its guarantees hold. We first present the tamper-evident logging construction, then describe the corresponding verification procedure, and finally define the adversary capabilities, trust assumptions, and protection boundaries relevant to forensic use.

A. TAMPER-EVIDENT DECISION LOGGING

Forensic decision records are useful only if their integrity and ordering can be verified after an incident. The framework therefore links the sequence

$$DR'_1, DR'_2, \dots, DR'_n$$

Algorithm 2 Tamper-Evident Hash-Chain Construction

Require: Sequential *DecisionRecord'* entries and trusted anchor h_0

Ensure: Tamper-evident decision log

```

1:  $h \leftarrow h_0$ 
2:
3: for each DecisionRecord'  $r_i$  do
4:    $r_i.\text{prev\_hash} \leftarrow h$ 
5:   Form  $\tilde{r}_i$  by excluding  $r_i.\text{record\_hash}$ 
6:    $b_i \leftarrow \text{UTF8}(\text{JSON}_{\text{canon}}(\tilde{r}_i))$ 
7:    $h \leftarrow \text{SHA-256}(b_i)$ 
8:    $r_i.\text{record\_hash} \leftarrow h$ 
9:   Append  $r_i$  to the decision log
10: end for
11:

```

using a cryptographic hash chain initialized from a trusted anchor h_0 .

For each record DR'_i , the integrity component is $I_i = \langle h_{i-1}, h_i \rangle$, where h_{i-1} is stored in `prev_hash`. The current digest is computed as

$$\begin{aligned} \widetilde{DR'_i} &= DR'_i \setminus \{\text{record_hash}\}, \\ b_i &= \text{UTF8}(\text{JSON}_{\text{canon}}(\widetilde{DR'_i})), \\ h_i &= \text{SHA-256}(b_i). \end{aligned}$$

The resulting h_i is stored in `record_hash`.

The implementation uses Python's `hashlib.sha256`. Canonical serialization uses compact JSON with lexicographically sorted object keys, preserved array order, `ensure_ascii=False`, and separators `(',', ':')`. The serialized representation is encoded directly as UTF-8 and contains no byte-order mark, indentation, insignificant whitespace, or trailing line terminator. The populated `prev_hash` is included in the hash input, whereas `record_hash` is excluded. The 32-byte digest is stored as a 64-character lowercase hexadecimal string. Algorithm 2 summarizes the corresponding construction procedure.

Any modification, interior deletion, or reordering that is not accompanied by valid recomputation produces a hash or predecessor-link inconsistency during verification. The mechanism provides tamper evidence rather than tamper prevention and depends on the trusted anchor or another independent integrity reference remaining uncompromised.

B. INTEGRITY VERIFICATION AND TAMPERING DETECTION

Given an extracted decision log and the trusted initial anchor h_0 , integrity verification proceeds by replaying the chain from the beginning and checking whether each record remains consistent with the stored linkage and content hash. Formally, verification computes

$$\hat{h}_i = H(DR'_i)$$

for each record in sequence. Let q_i denote the value stored in `temporal_metadata.sequence_number`.

Algorithm 3 Verification of the Decision-Record Hash Chain

Require: Ordered DecisionRecord' sequence $r_1, \dots, r_n$ and trusted anchor h_0

Ensure: Valid chain or detected inconsistency

```

1:  $h \leftarrow h_0$ 
2:  $q \leftarrow 1$ 
3:
4: for each record  $r_i$  in the extracted order do
5:    $q_i \leftarrow r_i.\text{temporal\_metadata.sequence\_number}$ 
6:   if  $q_i \neq q$  then
7:     Flag a sequence inconsistency and terminate
8:   end if
9:   if  $r_i.\text{prev\_hash} \neq h$  then
10:    Flag a predecessor-link inconsistency and terminate
11:   end if
12:   Form  $\tilde{r}_i$  by excluding  $r_i.\text{record\_hash}$ 
13:    $b_i \leftarrow \text{UTF8}(\text{JSON}_{\text{canon}}(\tilde{r}_i))$ 
14:    $\hat{h}_i \leftarrow \text{SHA-256}(b_i)$ 
15:   if  $r_i.\text{record\_hash} \neq \hat{h}_i$  then
16:    Flag a record-hash inconsistency and terminate
17:   end if
18:    $h \leftarrow \hat{h}_i$ 
19:    $q \leftarrow q + 1$ 
20: end for
21: Report the extracted chain as valid
22:

```

Verification requires

$$\begin{aligned}
 DR'_i.\text{prev_hash} &= h_{i-1}, \\
 q_i &= i, \\
 DR'_i.\text{record_hash} &= \hat{h}_i
 \end{aligned}$$

for all $i \in \{1, \dots, n\}$. A violation of any condition indicates a content, linkage, ordering, or deletion inconsistency in the extracted chain.

During verification, `record_hash` is excluded, the canonical UTF-8 representation is reconstructed, and the SHA-256 digest is recomputed independently. After a successful comparison, the recomputed digest becomes the expected `prev_hash` for the next record.

This procedure enables the detection of three principal forms of post-incident tampering. First, record *modification* changes the canonical record contents and therefore invalidates the stored hash. Second, record *deletion* breaks chain continuity and typically produces inconsistencies in both linkage and sequence progression. Third, record *reordering* alters predecessor relationships and invalidates the hash-chain structure even when the individual record contents remain unchanged.

Verification requires only the stored P2 records and the applicable trusted integrity references; omitted prompts, unrestricted reasoning, telemetry, and model states are not needed. Because retained records may still disclose contextual and selected-action details, the stored log requires appropriate access control.

Algorithm 3 summarizes the verification procedure.

Accordingly, local verification allows an investigator to test the extracted records for hash, predecessor-link,

and sequence consistency. Establishing whether an internally consistent log is complete and corresponds to the previously committed history additionally requires an uncompromised external checkpoint or corroborating copy.

C. THREAT MODEL, TRUST ASSUMPTIONS, AND PROTECTION BOUNDARY

The threat model considered in this work focuses on *post-incident* or *post-commitment* manipulation of forensic decision logs. The adversary is assumed to gain access to stored records after one or more DecisionRecord' entries have already been generated and committed to the log. Such an adversary may be an external attacker with access to stored files, a malicious insider, or compromised onboard software that attempts to alter retained forensic evidence after the fact.

Under this model, the adversary may read stored decision records, modify the contents of one or more records, delete records from the sequence, reorder records, splice log segments, or attempt to present an altered log as authentic during later forensic analysis. The proposed hash-chained logging mechanism is designed to make such post-hoc manipulation detectable through violations of chain continuity, record-hash consistency, and sequence progression.

However, the adversary is *not* assumed to control every trust anchor in the system. The framework therefore requires that at least one independent integrity reference remain uncompromised, such as a securely provisioned genesis value h_0 , periodically exported hash checkpoints, write-once or append-only storage regions, trusted offboard copies, or corroborating external records maintained by ground infrastructure. The exact realization of this trust anchor may vary across deployments, but the essential assumption is that an adversary cannot silently replace both the decision log and all independent reference points simultaneously. Importantly, the genesis value h_0 alone is sufficient to anchor chain verification only when records remain available; reliable detection of complete log deletion requires at least one additional trusted external reference, such as an exported checkpoint, offboard copy, or expected sequence summary.

The framework also has a clear scope boundary with respect to *runtime compromise*. It does not claim to prevent or fully detect manipulation of the decision-generation process before a record is committed. If an adversary alters the LLM, the record-construction routine, or the contextual inputs prior to logging, the resulting record may faithfully preserve a compromised decision. In that case, the mechanism can still protect the integrity of the stored record, but it cannot by itself establish that the underlying decision process was uncompromised. This distinction between *log integrity* and *decision integrity* is fundamental to the present design.

Under partial compromise, local verification detects uncomputed record modification, interior deletion, reordering,

and broken linkage. Coherently recomputed suffixes and complete-log or terminal-suffix deletion require an uncompromised external checkpoint or corroborating copy. If the adversary controls both the stored log and all external integrity references, the framework cannot establish which internally consistent history is authentic.

VI. EXPERIMENTAL SETUP

This section describes the experimental setup used to evaluate the proposed privacy-preserving forensic accountability framework. The goal of the experiments is not to assess UAV control quality or end-to-end autonomy, but to examine whether decision-level forensic evidence can be represented, protected, and verified under a privacy-constrained logging configuration. To this end, we describe the benchmark data used in the study, the decision-record generation pipeline, and the attack scenarios and evaluation metrics used to assess forensic integrity and utility.

A. DATASET DESCRIPTION

All experiments use the *UAVBench_MCQ 2K (Validated)* dataset [1], which contains 2,000 multiple-choice questions derived from UAV operational scenarios. The questions cover cyber-physical security, safety-critical decision making, policy compliance, navigation, and resource management.

UAVBench_MCQ was selected because the study evaluates forensic treatment of archived decision outputs rather than perception, flight control, or decision generation. The results therefore establish benchmark-level feasibility, not software-in-the-loop, hardware-in-the-loop, onboard, or closed-loop UAV validation; such integration is reserved for future work.

B. REPRODUCIBILITY AND IMPLEMENTATION DETAILS

The 2,000 UTF-8 JSON files in *UAVBench_API_200 0/scenario_mcqs/* were sorted by relative path. A local generator initialized with seed 42 then selected 200 files without replacement using `random.Random(42).sample(paths, 200)`. The selected paths were sorted again before processing, ensuring identical record order and sequence numbers across executions.

Each selected file produced one P2 record and one hash-chain entry. The implementation used Python 3.13.2; the dataset files were not modified, and no model training, fine-tuning, or parameter learning was performed. Serialization and hashing followed Section V-A.

Supplementary File S1 provides the ordered identifiers of the 200 selected instances and the SHA-256 digest of the manifest.

C. DECISION LOGGING PIPELINE

For each LLM-guided UAV decision event, the framework generates a privacy-aware forensic record, denoted *DecisionRecord'*, using the P2 profile defined in Section IV-B. The logging pipeline operationalizes the privacy-preserving

abstraction and integrity mechanism introduced in the preceding sections.

Each benchmark MCQ produces one P2 record. Fixed rules derive the event indicators from the normalized description, question, selected choice, and `style_id`. These indicators do not represent the model's internal rationale.

D. ATTACK SCENARIOS AND EVALUATION METRICS

Each integrity trial used a fresh copy of the valid 200-record chain. Twenty modification trials altered `selected_choice` at positions 5, 15, 25, ..., 195 without recomputing the hashes, while five deletion trials removed records at positions 25, 60, 100, 140, and 175.

Five additional trials modified records at positions 20, 60, 100, 140, and 180 and recomputed the complete affected suffix. These chains were tested by local verification and by comparison with the independently retained terminal checkpoint. Splicing, replay, suffix truncation, and checkpoint compromise are assessed analytically in Section VII-B.

Evaluation is based on the following metrics:

- **Attack-trial detection rate:** the proportion of independently executed attack trials in which verification reports an integrity or sequence inconsistency. The verifier reports the first failing record position rather than claiming to identify every alteration within a multiply corrupted chain.
- **Deletion and reordering detection:** the ability to detect missing or displaced records through hash-chain and sequence inconsistencies.
- **Storage overhead:** the average and total storage cost of logged *DecisionRecord'* entries under the P2 privacy profile.
- **Computational overhead:** the workstation-measured per-record latency for deterministic P2 construction, canonical serialization, SHA-256 computation, and in-memory chain-entry construction. These measurements do not establish onboard or resource-constrained UAV performance.

P2 is evaluated using three measures: (i) whole-field verbatim retention across the complete description, question, reference reason, and candidate-choice list; (ii) availability of seven predefined evidence groups; and (iii) average canonical JSON size reduction relative to the source MCQs. Retention does not capture partial, transformed, or inferential disclosure, while availability measures field presence rather than correctness or explanatory quality. Both size calculations use the same compact, sorted-key UTF-8 serialization.

No decision-accuracy metric is reported because the present study evaluates the deterministic forensic treatment of archived decision outputs rather than LLM reasoning correctness. The experiments examine record construction, evidence-field availability, bounded data minimization, integrity verification, storage, and workstation processing cost. They do not establish formal privacy, authenticated model identity, mission-level reconstruction, or onboard deployment performance.

All empirical evaluations used the same reproducibly selected 200-instance set and the archived outputs of meta-llama/llama-3.2-3b-instruct. A second dataset was not added because the study evaluates deterministic forensic logging rather than LLM generalization; broader dataset and mission-trace validation remains future work.

VII. EXPERIMENTAL RESULTS

This section presents the results of the experimental evaluation described in Section VI. We report the effectiveness of tamper and deletion detection, examine the degree of forensic investigability retained under the P2 privacy profile, and quantify the storage and computational overhead introduced by the proposed privacy-preserving logging mechanism.

A. TAMPERING AND DELETION DETECTION

All 200 intact P2 records passed local verification. Local verification detected all 20 single-record modification trials and all five interior-deletion trials, giving detection rates of 20/20 and 5/5, respectively.

All five coherently recomputed suffixes passed local verification but produced terminal-hash mismatches against the independently retained trusted checkpoint, resulting in 5/5 checkpoint-based detections. The verifier reports only the first detected inconsistency in each trial.

B. EXTENDED ADVERSARIAL-CASE ANALYSIS

The suffix-recomputation experiment showed that all five altered chains remained locally consistent but were detected through mismatch with the independently retained terminal checkpoint. Analytically, unrecomputed record splicing and replay disrupt predecessor hashes, sequence numbers, or both, whereas coherently recomputed variants require an uncompromised external integrity reference for detection. Similarly, terminal truncation may leave an internally valid prefix and therefore requires a trusted checkpoint, expected terminal sequence number, or corroborating copy.

The mechanism is therefore tamper-evident rather than tamper-proof. No numerical detection rate is claimed for the analytically assessed cases. If an attacker controls both the stored log and every external integrity reference, the framework cannot determine which internally consistent history is authentic.

C. INVESTIGABILITY ANALYSIS

An intact P2 chain preserves the selected action, retained context references, declared model metadata, benchmark reasoning style, event indicators, temporal position, and integrity metadata. The fields stored under rationale_structured are deterministic scenario- and output-derived indicators; they do not reveal the model's internal reasoning or constitute an authenticated explanation.

Across the 200 records, none of the 800 complete description, question, reference-reason, and candidate-list fields was retained verbatim, while all seven predefined evidence groups were present in every record (1,400/1,400).

TABLE 2. Storage and computational overhead of DecisionRecord logging under the P2 profile.

Metric	Value
Logged records	200
Measured executions	30
Warm-up executions	1
Average record size	1,130.470 bytes
Total log size	220.795 KiB
Mean serialization/abstraction time	0.555486 ms
Mean SHA-256 computation time	0.006798 ms
Mean end-to-end latency per record	0.563214 ms

TABLE 3. Ablation and logging-rate sensitivity results.

Condition	Storage result	Compute result
Complete P2	1,130.470 B/record	0.563214 ms/record
Without structured event indicators	807.055 B/record	–
Without integrity metadata	956.470 B/record	–
1 decision/min	66.238 KiB/h	0.000939%
1 decision/s	3.881 MiB/h	0.056321%
10 decisions/s	38.812 MiB/h	0.563214%

The average canonical payload decreased from 2,226.405 to 1,130.470 bytes, a 49.224% reduction. Nevertheless, 99 records contained a numerical value and 81 contained an operational quantity or unit. These results establish bounded data minimization and evidence availability, not anonymization, explanatory completeness, or removal of all partial, transformed, numerical, or inferential information.

D. STORAGE AND COMPUTATIONAL OVERHEAD

Table 2 summarizes the storage and host-side processing costs of constructing and hash-chaining the 200 P2 records.

Timing used one unreported warm-up followed by 30 independently initialized executions of the ordered 200-record set, each beginning from the fixed genesis anchor. The timed interval covered P2 construction, canonical serialization, SHA-256 computation, integrity-field assignment, and in-memory chain-entry construction.

Mean per-record latency was 0.555486 ± 0.058962 ms for serialization and abstraction, 0.006798 ± 0.000768 ms for SHA-256, and 0.563214 ± 0.059802 ms end to end. The corresponding 95% confidence intervals were [0.533469, 0.577503], [0.006511, 0.007084], and [0.540884, 0.585544] ms, respectively. Values are mean $\pm$ sample standard deviation; confidence intervals use the 30 execution-level means and a two-sided Student-*t* distribution with 29 degrees of freedom.

Measurements used Python 3.13.2 on 64-bit Windows 10 Pro (10.0.19045), an Intel Core i7-6950X at 3.00 GHz (10 cores and 20 threads), and 32 GB of RAM. The experiment used record version 1.0p, the archived output of meta-llama/llama-3.2-3b-instruct, and the fixed genesis anchor defined as the SHA-256 digest of UAVBENCH-P2-GENESIS-v1. These host-side

measurements do not represent onboard UAV performance bounds.

E. ABLATION AND SENSITIVITY ANALYSIS

Across the 200 evaluated records, canonical P2 sizes ranged from 1,040 to 1,275 bytes, with a median of 1,128 bytes. For the field-ablation analysis, individual evidence groups were removed from temporary record copies before canonical serialization. Removing the structured event indicators reduced the average size to 807.055 bytes, a 28.609% decrease, while removing the integrity metadata reduced it to 956.470 bytes, a 15.392% decrease. These storage ablations are not proposed as alternative operational profiles.

Table 3 reports the ablation results and projects storage and processing demand using the measured averages of 1,130.470 bytes and 0.563214 ms per record.

The projected storage demand ranges from 66.238 KiB/h at one decision per minute to 38.812 MiB/h at ten decisions per second, while record construction uses less than 0.564% of one CPU core. These are analytical projections for discrete decision events rather than deployment or telemetry measurements.

With lowercase hexadecimal JSON encoding, replacing SHA-256 with SHA-512 would add 128 bytes per record, increasing the average size to approximately 1,258.470 bytes, or 11.323%. HMAC-SHA-256 would preserve the digest size but require secret-key provisioning and protection. These alternatives were assessed analytically and were not executed.

Let N denote the number of stored records and B_i the canonical byte length of record i . Constructing and appending record i requires $O(B_i)$ time and is independent of the number of preceding records because only the previous digest is needed. Repository storage is $O(\sum_{i=1}^N B_i)$, while complete verification requires $O(N)$ time for bounded-size records and $O(1)$ auxiliary verifier state.

Using the measured averages, a 30-day mission producing one decision per minute would generate 43,200 records, occupy approximately 46.574 MiB, and require 24.331 s of cumulative record-construction time. At one decision per second, it would generate 2,592,000 records, occupy approximately 2.729 GiB, and require approximately 24.331 min. These are linear workstation-based projections that exclude persistent-storage, indexing, communication, LLM inference, and onboard-contention costs; full-repository verification at these scales was not measured.

VIII. DISCUSSION

A. COMPARATIVE ADVANCEMENT AND PRACTICAL IMPLICATIONS

The principal advancement of the proposed framework lies not in any single mechanism considered independently, but in the integration of three capabilities that are usually treated separately: decision-level forensic representation, explicit data minimization, and tamper-evident ordering. LLM-oriented UAV studies and benchmarks generally

emphasize decision quality, whereas conventional UAV forensics focuses on telemetry, commands, and platform artifacts. Privacy and cryptographic mechanisms, in turn, typically address disclosure reduction or integrity without defining a dedicated forensic representation for individual LLM-guided decisions. The proposed framework connects these areas by representing each discrete decision as a compact P2 record and binding the resulting records into a verifiable sequence.

The conceptual DecisionRecord schema defines the information categories required to preserve a decision event for later investigation. The evaluated adapter is narrower: it deterministically constructs P2 records from the UAVBench MCQ structure. This distinction is important because the conceptual schema may support other decision interfaces, whereas the reported implementation and measurements apply only to the benchmark-specific adapter.

The privacy results demonstrate a bounded trade-off rather than comprehensive privacy protection. P2 avoided whole-field verbatim retention of all 800 evaluated description, question, reference-reason, and candidate-list fields, while preserving all 1,400 predefined evidence-group instances and reducing average canonical payload size by 49.224%. However, the residual audit showed that selected actions may still retain numerical and operational details. P2 should therefore be understood as a data-minimization profile, not as anonymization, PII detection, resistance to inference, or a formal privacy guarantee.

The integrity results similarly demonstrate a defined protection boundary. Local verification detected all independently executed unrecomputed modifications and interior deletions. Coherently recomputed suffixes remained locally consistent and were detectable only through comparison with the independently retained terminal checkpoint. The contribution is therefore a tamper-evident logging mechanism under partial compromise, rather than a claim that hash chaining alone can establish authenticity after complete compromise of the log and every external integrity reference.

Operationally, the framework should be viewed as an accountability layer downstream of decision generation. It preserves the selected action, retained context references, declared model metadata, deterministic event indicators, temporal position, and integrity state of a recorded decision. These fields support bounded post-incident interpretation, but they do not validate the decision, prevent an unsafe action, expose hidden model reasoning, authenticate the declared model identity, or prove that the recorded context was truthful before capture.

Table 4 places this decision-level artifact in relation to conventional UAV logging. The comparison is functional rather than performance-based because the two approaches preserve different evidence and operate at substantially different logging rates.

The two forms of logging are complementary. Conventional telemetry remains necessary for reconstructing physical state

TABLE 4. Functional comparison of conventional UAV logging and the proposed P2 decision-level logging.

Comparison criterion	Conventional UAV telemetry/control logging	Proposed P2 DecisionRecord/ logging
Primary evidence	Vehicle state, sensor measurements, communication events, control commands, and actuator behavior.	A discrete LLM-guided decision record containing identity, selected context references, declared model metadata, selected action, temporal position, and integrity metadata.
Decision-event interpretation	May show the physical state or command associated with an event, but generally does not represent an LLM-guided decision as a dedicated forensic artifact.	Retains the selected action and bounded scenario- and output-derived constraint, risk, violation, and confidence-status indicators. These indicators are not the model's internal rationale.
Data-minimization treatment	May retain detailed measurements, trajectories, locations, and diagnostic content according to the platform configuration.	Omits specified source content, including complete contextual descriptions, questions, reference reasons, candidate-choice lists, unrestricted reasoning, telemetry streams, and precise locations. It does not provide anonymization or formal privacy guarantees.
Integrity support	Depends on protections supplied by the platform, storage system, log format, or forensic-acquisition procedure.	Applies deterministic canonicalization and SHA-256 hash chaining to each record, subject to the trusted-reference assumptions defined in the threat model.
Principal forensic use	Reconstruction of vehicle motion, system state, communication behavior, and executed commands.	Recovery and integrity verification of the recorded decision event, retained context, declared metadata, event indicators, and position in the ordered log.
Logging pattern	Usually periodic or high-frequency because telemetry and control variables change continuously.	Event-driven: one discrete decision produces one P2 record and one hash-chain entry.
Evaluated overhead	Not benchmarked in this study because retained signals, formats, and event rates differ across platforms.	An average of 1,130,470 bytes and 0.563214 ms of host-side in-memory construction time per record.

and vehicle execution, whereas P2 provides a dedicated artifact for the recorded decision event. Correlating both evidence sources through timestamps, sequence identifiers, and mission references could support richer reconstruction, but such cross-source correlation was not implemented in this study.

The conceptual schema is not tied to a particular UAV platform or model interface. Nevertheless, the evaluated adapter, privacy measurements, integrity experiments, and timing results are specific to the UAVBench MCQ representation and the reported host-side implementation. The findings do not establish software-in-the-loop, hardware-in-the-loop, onboard, edge-assisted, cloud-supported, or physical-UAV performance.

B. LIMITATIONS

The word *Validated* in *UAVBench_MCQ 2K (Validated)* is part of the official release name and is not treated as an independent certification of dataset quality. The release may contain duplicated scenario names, inconsistent option labels, answer-position imbalance, and semantically unusual combinations of environments and operating conditions. These issues may influence scenario realism and the indicators derived from scenario text, although they do not alter the deterministic serialization, hashing, sequence verification, or executed corruption tests. The evaluated instances are therefore identified through their filename-derived identifiers and the exact manifest supplied in Supplementary File S1.

The empirical evidence is benchmark-grounded rather than deployment-grounded. A collection of discrete MCQ instances does not reproduce continuous flight dynamics, noisy or conflicting sensor streams, communication delays, actuator uncertainty, rapidly changing environmental

conditions, or human intervention. Consequently, the evaluation establishes that the logging pipeline operates reproducibly on archived decision events; it does not establish mission-level effectiveness or suitability for a particular onboard resource envelope.

The framework also operates after a decision output has been captured. It cannot establish that the input context was truthful, that relevant events were not suppressed before record construction, that the declared model actually generated the output, or that the selected action was correct or safe. The deterministic indicators stored under *rationale_structured* summarize predefined textual conditions but do not represent causal reasoning, explanatory faithfulness, or authenticated model deliberation.

The privacy evaluation is deliberately bounded. Whole-field verbatim retention, evidence-group availability, canonical payload reduction, and the residual numerical-content audit do not measure semantic similarity, membership disclosure, linkage risk, partial quotation, or inference from combined fields. Likewise, the integrity guarantee depends on at least one uncompromised checkpoint or corroborating source. If an adversary controls record generation, the stored chain, and every external integrity reference, the framework cannot determine which internally consistent history is authentic.

C. FUTURE WORK

The first priority is deployment-oriented validation. Future work should integrate the logger with software-in-the-loop and hardware-in-the-loop UAV environments, followed by controlled physical-platform studies. Such experiments should evaluate persistent storage, communication delays, onboard resource contention, checkpoint export, event-loss

behavior, and correlation between P2 records, telemetry, commands, and mission-state evidence.

A second direction is stronger provenance and trust anchoring. Hardware-backed key storage, trusted platform modules, secure boot, remote attestation, append-only repositories, and independently administered checkpoints could strengthen confidence in record generation and retention. Model-output signing or authenticated inference services could also help distinguish declared model metadata from cryptographically supported model provenance.

Future privacy evaluation should move beyond field omission and payload size. Configurable profiles could be assessed using semantic-leakage analysis, re-identification or linkage tests, sensitivity-aware transformation rules, and mission-specific regulatory requirements. Such work could determine when retaining operational quantities is forensically necessary and when stronger generalization or redaction is justified.

Finally, broader evaluation is needed across additional UAV datasets, decision interfaces, model families, and autonomous cyber-physical domains. Applying the conceptual record to ground vehicles, maritime platforms, and robotic monitoring systems would help determine which components are UAV-specific and which form a more general pattern for accountable machine-generated decision logging.

IX. CONCLUSION

This paper introduced a privacy-preserving forensic accountability framework for LLM-driven UAV decision systems. By treating discrete LLM-guided decisions as first-class forensic artifacts, the framework preserves selected context, actions, declared model metadata, structured event indicators, temporal position, and integrity evidence for post-incident investigation. Privacy preservation is operationalized as bounded data minimization rather than anonymization, formal privacy protection, or resistance to inferential disclosure.

The framework formalizes a conceptual *DecisionRecord* and evaluates a deterministic UAVBench-specific P2 adapter. Across 200 sampled instances, none of the 800 evaluated complete source-field instances was retained verbatim, all 1,400 predefined evidence-group instances remained available, and the average canonical payload was reduced by 49.224%. Local verification detected all 20/20 record modifications and all 5/5 interior deletions. Five coherently recomputed suffixes remained locally consistent but were all detected using an independently retained trusted terminal checkpoint. Mean end-to-end record-construction latency was 0.563214 ms per record, and the complete chain occupied 220.795 KiB.

These findings establish benchmark-grounded feasibility for compact, data-minimized, and tamper-evident preservation of discrete LLM-guided UAV decisions. They do not establish decision accuracy, authenticated model identity, recovery of hidden reasoning, complete mission reconstruction, onboard performance, or protection when both

the stored log and every external integrity reference are compromised.

Future work will validate the framework in closed-loop and resource-constrained UAV environments, correlate decision records with telemetry and ground-station evidence, evaluate stronger privacy and leakage controls, and investigate hardware-backed trust anchoring and adapters for additional datasets and autonomous cyber-physical platforms.

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